1️⃣ Insurance Policy Industry

Title: *"Intelligent Insurance Policy & Claim Resolution Multi-Agent System"*

Context:
A large insurance company needs a next-generation **multi-agent AI assistant** that can **collaboratively** handle complex customer queries regarding policy details, premium calculations, claim eligibility, fraud detection, and regulatory compliance. Traditional chatbots fail because they can't dynamically coordinate multiple specialized agents for different aspects of the task.

Problem Statement:
You are required to **design and implement a multi-agent Agentic AI application** that:

1. **Uses only**:
    - **Gemini Free API Key** (Google Flash 1.5 or 2.5 model)
    - **CrewAI, LangGraph, LlamaIndex, AutoGen, and Semantic Kernel**
    - **Multi-agent communication protocols**: A2A (Agent-to-Agent) and MCP Server
    - **Agent personas** with **memory, tools, and iterative plans**
2. **Implements a multi-agent system** with specialized agents:
    - **Policy Expert Agent** – retrieves and explains policy clauses from indexed internal PDFs (via LlamaIndex).
    - **Claim Eligibility Agent** – checks customer details and determines eligibility using a mock API.
    - **Fraud Risk Assessment Agent** – runs fraud risk analysis using historical claim data.
    - **Regulatory Compliance Agent** – verifies claim against updated insurance regulations.
3. **Uses skill composition** in Semantic Kernel to combine small semantic functions (e.g., text classification, data extraction) into larger workflows.

4. **Implements a cognitive architecture** for **dynamic task resolution**—agents should decide which agent leads the conversation at each step.

5. **Supports multimodal interaction**—if a user uploads a claim form (PDF/Image), the system extracts and processes the data using an agent tool.

6. **Uses custom loops for agent chains** in both local and hybrid mode (local + API calls).

7. **Follows design patterns** for scaling the multi-agent solution.

8. **Runs long-running cooperative agent loops** for research-intensive queries (e.g., investigating past claims trends).

9. **Streamlit UI integration** for customer interaction and internal employee review.

10. **Deploys over Docker** and **hosts the complete solution on Google Cloud Platform** (Cloud Run or Compute Engine).

Expected Deliverable:

- Multi-agent **CrewAI + LangGraph + Semantic Kernel** workflow with cognitive task routing.

- **LlamaIndex-powered knowledge retrieval** from insurance policy documents.

- **Custom tool APIs** for claims and risk assessment.

- **Streamlit UI** for real-time conversation and file upload.

- **Docker + GCP deployment** ready.

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2 Finance / Fund Management Sector

Title: *"AI-Powered Multi-Agent Investment Portfolio & Risk Advisor"*

Context:
A wealth management company wants an advanced **multi-agent AI system** to help financial advisors and clients with personalized investment strategies, portfolio rebalancing, risk forecasting, and regulatory checks. The goal is to create a **cooperative agent ecosystem** that

can continuously refine investment plans based on market data, user goals, and compliance rules.

Problem Statement:
You are required to **design and implement a multi-agent financial advisor** that:

1. **Uses only**:
    - **Gemini Free API Key** (Google Flash 1.5 or 2.5 model)
    - **CrewAI, LangGraph, LlamaIndex, AutoGen, and Semantic Kernel**
    - **Multi-agent communication protocols**: A2A (Agent-to-Agent) and MCP Server
    - **Agent personas** with **memory, tools, and iterative plans**
2. **Implements a multi-agent system** with specialized agents:
    - **Market Analyst Agent** – fetches and interprets live market data via API.
    - **Portfolio Manager Agent** – optimizes asset allocation based on user profile and market conditions.
    - **Risk Assessment Agent** – runs Monte Carlo simulations and volatility checks.
    - **Regulatory Compliance Agent** – checks portfolio changes against local/international investment laws.
3. **Uses skill composition** in Semantic Kernel to chain calculations, trend analysis, and report generation.
4. **Builds a cognitive architecture** where agents negotiate and refine investment strategies dynamically.
5. **Supports multimodal interactions**—if a user uploads their investment portfolio as a CSV, the system parses and processes it via a tool-enabled agent.
6. **Uses custom loops for agent chains** in local and hybrid mode (local AI + cloud APIs).
7. **Implements design patterns** for scaling financial multi-agent workflows.
8. **Supports long-running cooperative loops** for deep market research tasks (e.g., quarterly analysis).

9. **Streamlit UI integration** for interactive portfolio visualization and strategy simulation.

10. **Deploys over Docker** and **hosts the solution on Google Cloud Platform** (Cloud Run or Compute Engine).

Expected Deliverable:

- Multi-agent **CrewAI + LangGraph + Semantic Kernel** architecture with financial reasoning.
- **LlamaIndex integration** for querying compliance guidelines and investment reports.
- **Custom APIs** for market data and simulation.
- **Streamlit dashboard** for visualization, interaction, and file uploads.
- **Docker + GCP deployment** ready.